higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MATHEMATICS N5

27 March 2026

This marking guideline consists of 11 pages.

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TVET Exam Papers NATED & NCV	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
N4-N6 Building Administration NATED	**N4-N6** Building& Structural Surveying NATED	**N4-N6** Building & Structural Construction NATED	**N1-N3** Building Science NATED	**N1-N3** Building Drawing NATED	**N4** Mechanical Draughting NATED
N1-N3 Electrical Trade Theory NATED	**N2-N3** Diesel Trade Theory NATED	**N5-N6** Strength of Materials & Structures NATED	**N3** Building & Civil Technology NATED	**N1-N2** Fitting and Machining Theory NATED	**N1-N4** Plating and Structural Steel Drawing NATED
N1-N3 Motor Trade Theory NATED	**N5-N6** Mechanical Drawing & Design NATED	**N1-N2** Plumbing Theory NATED	**N4-N6** Fault Finding & Protective Devices NATED	**N2-N3** Instrument Trade Theory NATED	**N5-N6** Engineering Physics NATED
N5-N6 Fluid Mechanics NATED	**N4-N6** Supervisory Management NATED	**N4-N6** Digital Electronics NATED	**N6** Control Systems NATED	**N1-N2** Bricklaying &Plastering Theory NATED	More…

PRINCIPLES RELATED TO THE MARKING OF MATHEMATICS

1. If more information is given than marks allocated:

Stop marking when the maximum marks are reached and draw a wavy line with the word **'max'** in the right-hand margin.

Draw a line through the remaining work to indicate to the examination assistants that the entire answer has been considered.

2. Overlapping/carry-over questions

Where an incorrect answer could be carried over to the next step, the first answer will be deemed incorrect. However, should the incorrect answer be carried over correctly, the marker must re-calculate the values using the incorrect answer from the first calculation. If correctly used, the candidate should receive full marks for subsequent calculations.

3. Incorrect numbering of asnwers

If a candidate numbered an answer incorrectly, but the answer is in the correct position according to the sequence of the questions in the question paper, circle the incorrect numbering and credit marks for the response.

If questions were answered randomly not following the same sequence as in the question paper and the candidate numbered incorrectly, the response should **NOT** be marked.

4. Criteria/Rubric for marking a graph

X-axis and Y-axis: Correctly calibrated and labelled, and accuracy (80%+ correctly plotted)
Turning points/point of inflection clearly indicated

5. Be sensitive to the meaning of an answer, which may have been approached/stated in a different way.

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QUESTION 1

1.1 1.1.1

$$\log_3 y = \lim_{x \to 0} \frac{1-\cos^2 x}{x^2} \quad : \frac{0}{0}$$

$$= \lim_{x \to 0} \frac{2\sin x \cos x}{2x}$$

$$= \lim_{x \to 0} \frac{2(\cos^2 x - \sin^2 x)}{2}$$

$$= 1$$

(3)

Alternative

$$= \lim_{x \to 0} \frac{2\sin x \cos x}{2x}$$

$$= \lim_{x \to 0} \frac{\sin 2x}{2x}$$

$$= \lim_{x \to 0} \frac{2\cos 2x}{2}$$

$$= 1$$

1.1.2 $y = 3^1 = 3$ (1)

1.2 $f(x) = \frac{\sin x}{x+3}$

$= (-\infty;\ -3) \cup (-3;\ \infty)$ or $x \neq -3; x \in R$ (2)

[6]

QUESTION 2

2.1

$$f'(x) = \lim_{h \to 0} \frac{\frac{1-(x+h)}{(x+h)+2} - \frac{1-x}{x+2}}{h}$$

$$= \lim_{h \to 0} \frac{\frac{(1-x-h)(x+2)-(1-x)(x+h+2)}{(x+h+2)(x+2)}}{h}$$

$$= \lim_{h \to 0} \frac{\frac{x-x^2-xh+2-2x-2h-x-h-2+x^2+xh+2x}{(x+h+2)(x+2)}}{h}$$

$$= \lim_{h \to 0} \frac{\frac{-3h}{(x+h+2)(x+2)}}{h}$$

$$= \lim_{h \to 0} \frac{-3}{(x+h+2)(x+2)}$$

$$= -\frac{-3}{(x-2)^2}$$

(5)

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2.2 2.2.1

$$y=\left(x^2+5\right)^{-\frac{3}{2}}\left(sec\,5x^2\right)$$

$$\frac{dy}{dx}=-\frac{3}{2}\left(x^2+5\right)^{-\frac{5}{2}}\times 2x\times sec\,5x^2+\left(x^2+5\right)^{-\frac{3}{2}}\times\left(sec\,5x^2\tan 5x^2\right)\times 10x$$ (4)

2.2.2

$$y=\frac{1-sec\left(\theta^2x\right)}{\theta^3x^3-\cot\theta}$$

$$\frac{dy}{dx}=\frac{\left(\theta^3x^3-\cot\theta\right)\left[sec\left(\theta^2x\right)\tan\left(\theta^2x\right)\right]\times\theta^2-¿\left[1-sec\left(\theta^2x\right)\right]\left(3\theta^3x^2+cosec^2x\right)}{\left(\theta^3x^3-\cot\theta\right)^2}$$ (4)

2.3 $$y=\left[\ln\left(x^2+7\right)\right]^{\cos^{-1}x}$$

$$\ln y=\cos^{-1}x\ln\left[\ln\left(x^2+7\right)\right]$$

$$\frac{1}{y}\frac{dy}{dx}=\cos^{-1}x\times\frac{1}{\ln\left(x^2+7\right)}\times\frac{1}{x^2+7}\times 2x+\left(-\frac{1}{\sqrt{1-x^2}}\right)\times\ln\left[\ln\left(x^2+7\right)\right]❑$$

$$\frac{dy}{dx}=y\left[\frac{2x\cos^{-1}x}{\left(x^2+7\right)\ln\left(x^2+7\right)}-\frac{\ln\left[\ln\left(x^2+7\right)\right]}{\sqrt{1-x^2}}\right]$$

$$=\left[\ln\left(x^2+7\right)\right]^{\cos^{-1}x}\left[\frac{2x\cos^{-1}x}{\left(x^2+7\right)\ln\left(x^2+7\right)}-\frac{\ln\left[\ln\left(x^2+7\right)\right]}{\sqrt{1-x^2}}\right]$$ (4)

2.3 $$\cos\left(x^2y^3\right)+3x=y\cot 2x$$

$$-\sin\left(x^2y^3\right)\times\left(2xy^3+3x^2y^2\frac{dy}{dx}\right)+3=\cot 2x\frac{dy}{dx}+y\times\left(-2\,cosec^2 2x\right)$$

$$-2xy^3\sin\left(x^2y^3\right)-3x^2y^2\sin\left(x^2y^3\right)\frac{dy}{dx}+3=\cot 2x\frac{dy}{dx}-2y\,cosec^2 2x$$

$$\left[-3x^2y^2\sin\left(x^2y^3\right)-\cot 2x\right]\frac{dy}{dx}=2xy^3\sin\left(x^2y^3\right)-2y\,cosec^2 2x$$

$$\frac{dy}{dx}=\frac{2xy^3\sin\left(x^2y^3\right)-2y\,cosec^2 2x}{-3x^2y^2\sin\left(x^2y^3\right)-\cot 2x}$$ (5)

[22]

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QUESTION 3

3.1 3.1.1 $f(x)=x^3+x^2-x-5$

$f'(x)=3x^2+2x-1=0$

$(x+1)(3x-1)=0$

$x=-1 \quad \vee \quad x=\frac{1}{3} \quad = 0{,}333$

$y=-4 \qquad y=-\frac{140}{27} \quad =-5{,}18$

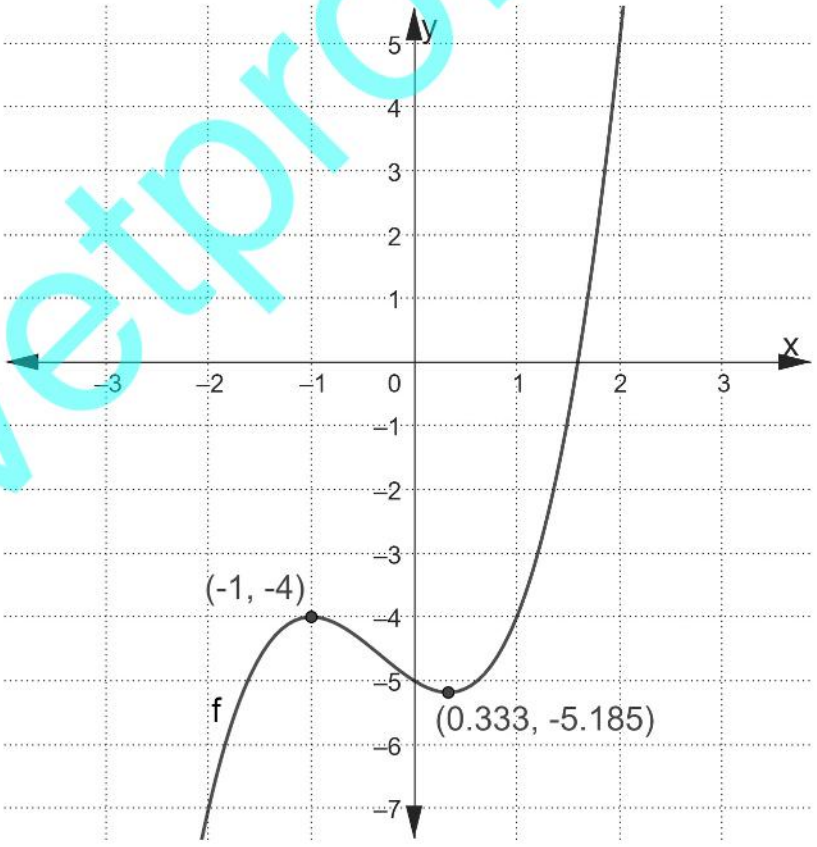
Thus, the coordinates of the turning points are $(-1;\ -4)$ and $\left(\frac{1}{3};-\frac{140}{27}\right)$ (3)

3.1.2

x	-2	-1	0	1	2
y	-7	-4	-5	-4	5

= 1 mark for every 2 correct answers (2)

3.1.3



1 mark for the shape
1 mark for the indication of the turnings point on the graph (2)

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3.1.4 *Let* $x_0 = 1,5$

$f(1,5) = -0,875$

$f'(1,5) = 8,75$

$x_1 = 1,5 - \frac{-0,875}{8,75}$

$= 1,6$

$x_2 = 1,6 - \frac{0,056}{9,88}$

$= 1,594$ (4)

3.2 $V = l^2 h$

$100 = l^2 h$

$\therefore h = \frac{100}{l^2}$

$A = 4lh + l^2$

$A = 4l\left(\frac{100}{l^2}\right) + l^2$

$= \frac{400}{l} + l^2$

$A' = 2l - \frac{400}{l^2} = 0$

$2l^3 - 400 = 0$

$l^3 - 200 = 0$

$l = 5,848 \ cm \quad \wedge \quad h = 2,924 \ cm$ (5)

3.2 $A = 4\pi r^2$

$\frac{dA}{dt} = 8\pi r \frac{dr}{dt}$

$= 8\pi(10 \ cm)(0,1 \ cm/s)$

$= 8\pi \ cm^2/s = 25,133 \ cm^2/s$ (3)

[19]

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QUESTION 4

4.1 4.1.1

$$\int \frac{x}{\sqrt{1-25x^4}}\,dx$$

$$=\int \frac{x}{\sqrt{1-\left(5x^2\right)^2}}\,dx$$

$let\ \ u=5x^2$

$$\frac{1}{10}\,du = x\,dx$$

$$=\frac{1}{10}\int \frac{1}{\sqrt{1-u^2}}\,du$$

$$=\frac{1}{10}\left[\sin^{-1}u\right]+C$$

$$=\frac{1}{10}\sin^{-1}5x^2+C$$

(3)

4.1.2 $$\int \sin^3 x\cos^4 x\,dx$$

$$=\int \sin^2 x\sin x\cos^4 x\,dx$$

$$\int \left(1-\cos^2 x\right)\sin x\cos^4 x\,dx$$

$u=\cos x$
$du=-\sin x\,dx$

$$=-\int \left(1-u^2\right)u^4\,du$$

$$=-\int \left(u^4-u^6\right)du$$

$$=-\frac{1}{5}u^5-\frac{1}{7}u^7+C$$

$$=-\frac{1}{5}\cos^5 x-\frac{1}{7}\cos^7 x+C$$

(5)

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4.1.3 $\int \frac{2x^3+x^2-4x+1}{x-2}dx$

$=\int\left(2x^2+5x+6+\frac{13}{x-2}\right)dx$

$=\frac{2}{3}x^3+\frac{5}{2}x^2+6x+13\ln(x-2)+C$ (5)

4.1.4 $\int xe^{4x}dx$

let $f(x)=x$ $g'(x)=e^{4x}dx$

$f'(x)=1\,dx$ $g(x)=\frac{1}{4}e^{4x}$

$\int xe^{4x}dx=x\times\frac{1}{4}e^{4x}-\int\frac{1}{4}e^{4x}dx$

$=\frac{xe^{4x}}{4}-\frac{1}{16}e^{4x}+C$ (4)

4.2 $\int\frac{2x^2-10}{(x-4)^3}dx$

$\frac{2x^2-10}{(x-4)^3}=\frac{A}{x-4}+\frac{B}{(x-4)^2}+\frac{C}{(x-4)^3}$

$2x^2-10=A(x-4)^2+B(x-4)+C$

let $x=4$; $C=22$

$2x^2-10=A(x^2-8x+16)+B(x-4)+C$

$2x^2-10=Ax^2+(B-8A)x+(16A-4B+C)$

$A=2$

$B-8A=0$

$16A-4B+C=-10$

$B=16$

$\int\left(\frac{2}{x-4}+\frac{16}{(x-4)^2}+\frac{22}{(x-4)^3}\right)dx$

$=2\ln(x-4)-\frac{16}{x-4}-\frac{11}{(x-4)^2}+C$ (7)

[24]

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QUESTION 5

5.1

$$\int_0^{\infty} \frac{1}{\sqrt{3-x}}\,dx$$

$$= \lim_{t\to\infty} \int_0^{t} \frac{1}{\sqrt{3-x}}\,dx$$

$$= \lim_{t\to\infty} \left[-2\sqrt{3-x}\right]_0^{t}$$

$$= \lim_{t\to-\infty} \left[-2\sqrt{3-t} + 2\sqrt{3}\right]$$

$$= \infty + 2\sqrt{3}$$

$$= \infty$$

ALTERNATVE

$$= \int_0^{\infty} \frac{1}{\sqrt{3-x}}\,dx$$

$$= \left[-2\sqrt{3-x}\right]_0^{t}$$

$$= \left[-2\sqrt{3-t} + 2\sqrt{3}\right]$$

$$= \infty + 2\sqrt{3}$$

$$= \infty$$

(4)

5.2 5.2.1

$$10 - x = -x^2 + 10x - 8$$

$$x^2 - 11x + 18 = 0$$

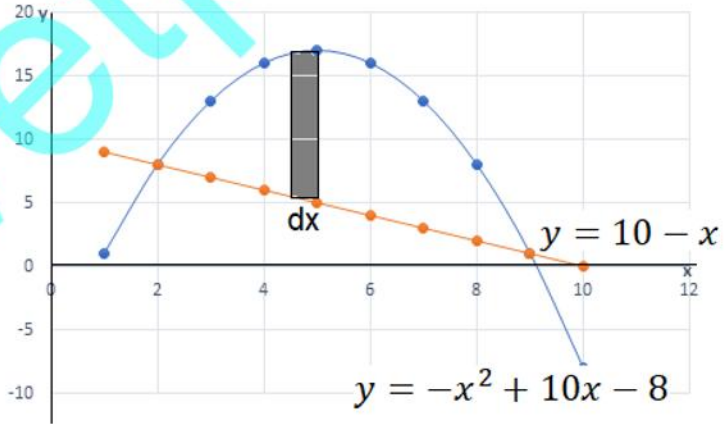
$$(x-2)(x-9) = 0$$

$$x = 2 \quad \vee \quad x = 9$$

$$y = 8 \quad \vee \quad y = 1$$

Thus, the coordinates of the points of intersection are $(2;\ 8)$ and $(9;1)$ (3)

5.2.2



= 1 mark for the indication of each enclosed area

= 1 mark for the indication of the vertical strip (2)

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5.2.3 $$A=\int_a^b (y_1-y_2)\,dx$$

$$=\int_2^9 \left[(-x^2+10x-8)-(10-x)\right]dx$$

$$=\int_2^9 (-x^2+11x-18)\,dx$$

$$=\left[-\frac{1}{3}x^3+\frac{11}{2}x^2-18x\right]_2^9$$

$$=57{,}167\ \ units^2 \ \vee\ \frac{343}{6}units^2$$ (4)

5.2.4 $$V=\pi\int_a^b (y_1^{\,2}-y_2^{\,2})\,dx$$

$$=\pi\int_2^9 \left[(-x^2+10x-8)^2-(10-x)^2\right]dx$$

$$=\pi\int_2^9 (x^4-20x^3+115x^2-140x-36)\,dx$$

$$=\pi\left[\frac{1}{5}x^5-5x^4+\frac{115}{3}x^3-70x^2-36x\right]_2^9$$

$$=\frac{16121}{15}\pi\ \ units^3 \quad \vee \quad =3376{,}374\ \ units^3$$ (5)

5.3 $$dI_z=r^2\,dA$$

$$=x^2\cdot 2\pi x\ \ dx$$

$$I=2\pi\int_0^7 x^3\ \ dx$$

$$=2\pi\left[\frac{x^4}{4}\right]_0^7$$

$$=\frac{2401\pi}{2}cm^4 \quad \vee\ =3771.482\ \ cm^4$$ (4)

[22]

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QUESTION 6

6.1 $x^2 e^y \frac{dy}{dx} = 1$; $y = 0$ *when* $x = 1$

$$e^y\,dy = \frac{1}{x^2}\,dx$$

$$e^y = -\frac{1}{x} + C$$

$$e^0 = -\frac{1}{1} + C$$

$$C = 2$$

$$e^y = -\frac{1}{x} + 2$$ (4)

6.2 $$e^{-x}\frac{d^2y}{dx^2} = e^{4x} + e^{-x}\sin 3x$$

$$\frac{d^2y}{dx^2} = e^{5x} + \sin 3x$$

$$\frac{dy}{dx} = \frac{1}{5}e^{5x} - \frac{1}{3}\cos 3x + A$$

$$y = \frac{1}{25}e^{5x} - \frac{1}{9}\sin 3x + Ax + B$$ (3)

[7]

TOTAL: 100

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